

TECHNICAL DATA / EXTRA INFORMATION

Pulley

*Generic technical datasheet for mechanical theatre equipment compliance records***Important scope note**

This document is a generic technical datasheet template for a pulley, sheave block or diverter pulley used in theatre, studio or entertainment lifting and suspension systems. It is not a substitute for the manufacturer manual, design calculations, rope system drawings, site risk assessment, inspection report or competent person assessment.

1. Equipment overview

A pulley or sheave is used to change the direction, path or mechanical advantage of a rope, wire rope or similar lifting medium. In theatre environments, pulleys may be used in counterweight systems, rope sets, winch systems, flying systems, diverter arrangements, curtain systems or other overhead suspension arrangements.

The pulley should be considered as part of the complete load path. This includes the pulley body, sheave, axle, bearings or bushes, side plates, attachment point, fixings, rope or wire rope, terminations, supporting structure and any related lifting machinery.

Summary	
Typical use	Changing the direction of fibre rope, wire rope, belt or line within a lifting, suspension, flying or movement system.
Primary risk areas	Overhead load, rope wear, sheave seizure, axle wear, side loading, incorrect rope size, groove damage, poor fixing and overload.
Portal equipment class	Lifting accessory / load-path component.
Not covered by this template	Manufacturer-specific load ratings, rope groove dimensions, bearing data, system calculations or inspection values unless added by a competent person.

2. Identification and site details

Asset record

Venue / site	To be completed
Area / room / stage	To be completed
Equipment ID / asset number	To be completed
Pulley location / line number	To be completed
Manufacturer	To be completed
Model / type	To be completed
Serial number / batch ID	To be completed
Installed by	To be completed
Date installed / last replaced	To be completed
Related drawing / system reference	To be completed

3. Technical data to confirm

Technical information	
Safe Working Load / WLL	To be confirmed from manufacturer information, system design and current inspection records.
Pulley type	Single sheave / double sheave / head block / loft block / diverter / snatch block / other - to be completed.
Sheave diameter	To be completed. Confirm suitability for the rope or wire rope diameter and construction.
Rope / wire rope diameter	To be completed
Rope type and construction	Fibre rope / wire rope / belt / other - to be completed.
Groove profile	To be confirmed as suitable for the rope or wire rope in use.
Bearing / bush type	To be completed
Attachment method	Fixed axle, clevis, bracket, beam fitting, shackle, pin, bolted plate or other - to be completed.
Load direction / fleet angle	To be completed. Confirm that side loading and excessive fleet angle are avoided.

Inspection interval	To follow LOLER/PUWER requirements, written scheme, manufacturer instructions and site risk assessment.
Known limitations	To be completed - include loading, rope size, angle, movement, environment or access restrictions.

Compatibility note

Pulley suitability depends on the complete rope system. The sheave diameter, groove profile, rope diameter, rope construction, load angle, axle support and fixing method must all be compatible. A pulley with an adequate WLL may still be unsuitable if it damages the rope or introduces side loading.

4. Applicable compliance context

LOLER: A pulley used within a lifting arrangement or as a lifting accessory/load-path component should be treated as part of the lifting equipment. It should be strong and stable, suitable for the task, marked or traceable where appropriate, and examined or inspected by a competent person as required.

PUWER: The pulley is work equipment and should be suitable for its intended use, maintained in a safe condition and inspected where deterioration could create risk.

BS EN 17206: Where the pulley forms part of entertainment machinery or stage machinery, BS EN 17206 should be considered for load-bearing components, verification, inspection, documentation and prevention of uncontrolled movement.

ABTT Technical Standards: ABTT guidance supports recognised theatre-industry good practice for inspection, maintenance, documentation and competent operation in places of entertainment.

5. Installation and support considerations

- Confirm that the pulley is correctly rated for the load, rope type, rope diameter and direction of loading.
- Confirm that the supporting structure, bracket, beam, fixing plate, axle or suspension point is suitable for the imposed load.
- Check that the rope enters and leaves the sheave correctly and does not rub on side plates, guards, brackets or adjacent components.
- Avoid side loading, excessive fleet angle, poor rope alignment and any arrangement that forces the rope out of the sheave groove.
- Confirm that axles, pins, retaining clips, nuts, split pins or other retention devices are correctly fitted and secured.
- Where guards or covers are fitted, confirm that they prevent rope escape without trapping or damaging the rope.

6. Operating considerations

- Only use the pulley within the known SWL/WLL and system design limits.

- Do not use a pulley if the sheave does not rotate freely or if there is abnormal noise, vibration, wobble or binding.
- Do not allow the rope or wire rope to run over sharp edges, damaged grooves or seized sheaves.
- Do not use a pulley with an unknown rating, missing identification or incompatible rope size.
- Stop use immediately if rope tracking is poor, the rope jumps the groove, the sheave rubs the side plate or the pulley becomes hot in use.
- Do not modify, drill, weld, grind or repair a pulley unless assessed and approved by a competent person.

7. Inspection checklist

Typical checks

No.	Check item	Status / notes
1	Confirm identification, rating, rope size and intended use are known or otherwise traceable.	☐ OK ☐ Defect ☐ N/A Notes:
2	Inspect pulley body, cheeks, side plates, clevis, brackets and attachment points for cracks, distortion, corrosion or impact damage.	☐ OK ☐ Defect ☐ N/A Notes:
3	Inspect the sheave for cracks, worn grooves, sharp edges, flat spots, corrosion, damage or contamination.	☐ OK ☐ Defect ☐ N/A Notes:
4	Check that the sheave rotates freely without excessive play, roughness, binding, wobble or abnormal noise.	☐ OK ☐ Defect ☐ N/A Notes:
5	Inspect axle, pin, spindle, bearings, bushes, circlips, split pins, nuts and retaining devices for wear, looseness, damage or missing parts.	☐ OK ☐ Defect ☐ N/A Notes:
6	Check the rope or wire rope path through the pulley for correct alignment and absence of rubbing or side loading.	☐ OK ☐ Defect ☐ N/A Notes:

7	Inspect the rope or wire rope near the pulley for broken wires, abrasion, flattening, kinks, glazing, contamination or local wear.	☐ OK ☐ Defect ☐ N/A Notes:
8	Check that guards, covers or rope-retaining devices are present and correctly fitted where required.	☐ OK ☐ Defect ☐ N/A Notes:
9	Inspect supporting brackets, beams, fixings, suspension points and surrounding structure for movement, corrosion, cracking or distortion.	☐ OK ☐ Defect ☐ N/A Notes:
10	Check for evidence of overload, heat, seizure, poor lubrication, unusual wear pattern or unauthorised modification.	☐ OK ☐ Defect ☐ N/A Notes:
11	Operate the system where safe and confirm smooth running without snatching, rubbing, jump, drift or abnormal noise.	☐ OK ☐ Defect ☐ N/A Notes:
12	Confirm that inspection reports, maintenance records and any defect records are available and current.	☐ OK ☐ Defect ☐ N/A Notes:

8. Defects requiring action

- Missing or uncertain WLL/SWL, identification, rope size compatibility or inspection status.
- Cracked, bent, distorted, heavily corroded or damaged pulley body, cheek plates, brackets or attachment points.
- Sheave seized, rough-running, badly worn, cracked, loose, misaligned or with sharp/worn grooves.
- Damaged, missing or incorrectly fitted axle, pin, circlip, split pin, retaining nut or locking device.
- Evidence that the rope has been rubbing on side plates, guards, brackets or other fixed parts.
- Broken wires, kinks, flattening, severe abrasion, glazing or local rope damage associated with the pulley.
- Side loading, excessive fleet angle, rope jumping the groove or unstable rope tracking.
- Loose fixings, movement in support brackets or signs of structural distress.
- Unauthorised modification, non-standard repair, welding, drilling, grinding or replacement parts of unknown origin.

Defect response

If a pulley defect could affect safe use, remove the associated system from service, prevent further operation and refer it to a competent person. Do not return the system to service until the defect has been assessed, corrected and recorded.

9. Records and document links

Associated records	
Manufacturer datasheet / manual	To be attached or referenced
System drawing / rope path drawing	To be attached or referenced
Original design / calculations	To be attached or referenced where available
LOLER thorough examination report	To be attached or referenced
PUWER inspection / maintenance record	To be attached or referenced
Rope or wire rope certificate	To be attached or referenced where applicable
Defect / repair history	To be attached or referenced
Site-specific safe system of work	To be attached or referenced

10. Portal summary wording

Suggested portal note

Use this datasheet as a generic technical record for a pulley, sheave or diverter used in a theatre lifting or suspension system. Add manufacturer-specific data, rope compatibility, WLL/SWL, system drawings, inspection records and any site limitations before relying on it for inspection or maintenance planning.

Document status: generic editable template - complete site-specific details before use



User Manual

www.doughty-engineering.co.uk

Pulleys



Original Instructions

Revision 2 06/10/15

www.doughty-engineering.co.uk

The Doughty Pulley collection is a range of Pulleys designed for use within the theatre, film and TV Industry. The range is split onto two groups Standard Pulleys for loads up to 150Kg and Heavy Duty Pulleys for loads up to 600Kg.

Heavy Duty Pulleys are fitted with ball raced bearings, Standard Pulley have a Nylon sheave running on a plain shaft.

Doughty Pulleys are available in the following specifications:

Pt No	Type	Wire/Rope	Max Load	Dia	Width	Weight
Standard Pulleys (Wire)						
T42800	Single	4mm Wire	150Kg	100mm	22mm	1.00Kg
T42801	Double	4mm Wire	150Kg	100mm	53mm	1.60Kg
T42812	Triple	4mm Wire	150Kg	100mm	85mm	2.40Kg
T42813	Quad	4mm Wire	150Kg	100mm	116mm	3.10Kg
T42814	Quin	4mm Wire	150Kg	100mm	148mm	3.90Kg
T42820	Single	6mm Wire	150Kg	150mm	22mm	1.60Kg
T42821	Double	6mm Wire	150Kg	150mm	53mm	3.10Kg
T42822	Triple	6mm Wire	150Kg	150mm	85mm	4.60Kg
T42823	Quad	6mm Wire	150Kg	150mm	116mm	6.00Kg
T42824	Quin	6mm Wire	150Kg	150mm	148mm	7.60Kg
T42830	Single	6mm Wire	150Kg	180mm	22mm	2.00Kg
T42831	Double	6mm Wire	150Kg	180mm	53mm	3.90Kg
T42832	Triple	6mm Wire	150Kg	180mm	85mm	5.80Kg
T42833	Quad	6mm Wire	150Kg	180mm	116mm	7.70Kg
T42834	Quin	6mm Wire	150Kg	180mm	148mm	9.60Kg
Standard Pulleys (Rope)						
T42836	Single	12mm Rope	150Kg	100mm	22mm	1.00Kg
T42837	Double	12mm Rope	150Kg	100mm	53mm	1.50Kg
T42838	Triple	12mm Rope	150Kg	100mm	85mm	2.20Kg
T42839	Quad	12mm Rope	150Kg	100mm	116mm	2.90Kg
T42840	Quin	12mm Rope	150Kg	100mm	148mm	3.50Kg
T42846	Single	18mm Rope	150Kg	150mm	22mm	10.50Kg
T42847	Double	18mm Rope	150Kg	150mm	53mm	2.90Kg
T42848	Triple	18mm Rope	150Kg	150mm	85mm	4.30Kg
T42849	Quad	18mm Rope	150Kg	150mm	116mm	5.70Kg
T42850	Quin	18mm Rope	150Kg	150mm	148mm	7.00Kg
T42855	Single	18mm Rope	150Kg	180mm	22mm	1.90Kg
T42856	Double	18mm Rope	150Kg	180mm	53mm	3.70Kg
T42857	Triple	18mm Rope	150Kg	180mm	85mm	5.50Kg
T42858	Quad	18mm Rope	150Kg	180mm	116mm	7.40Kg
T42859	Quin	18mm Rope	150Kg	180mm	148mm	9.20Kg

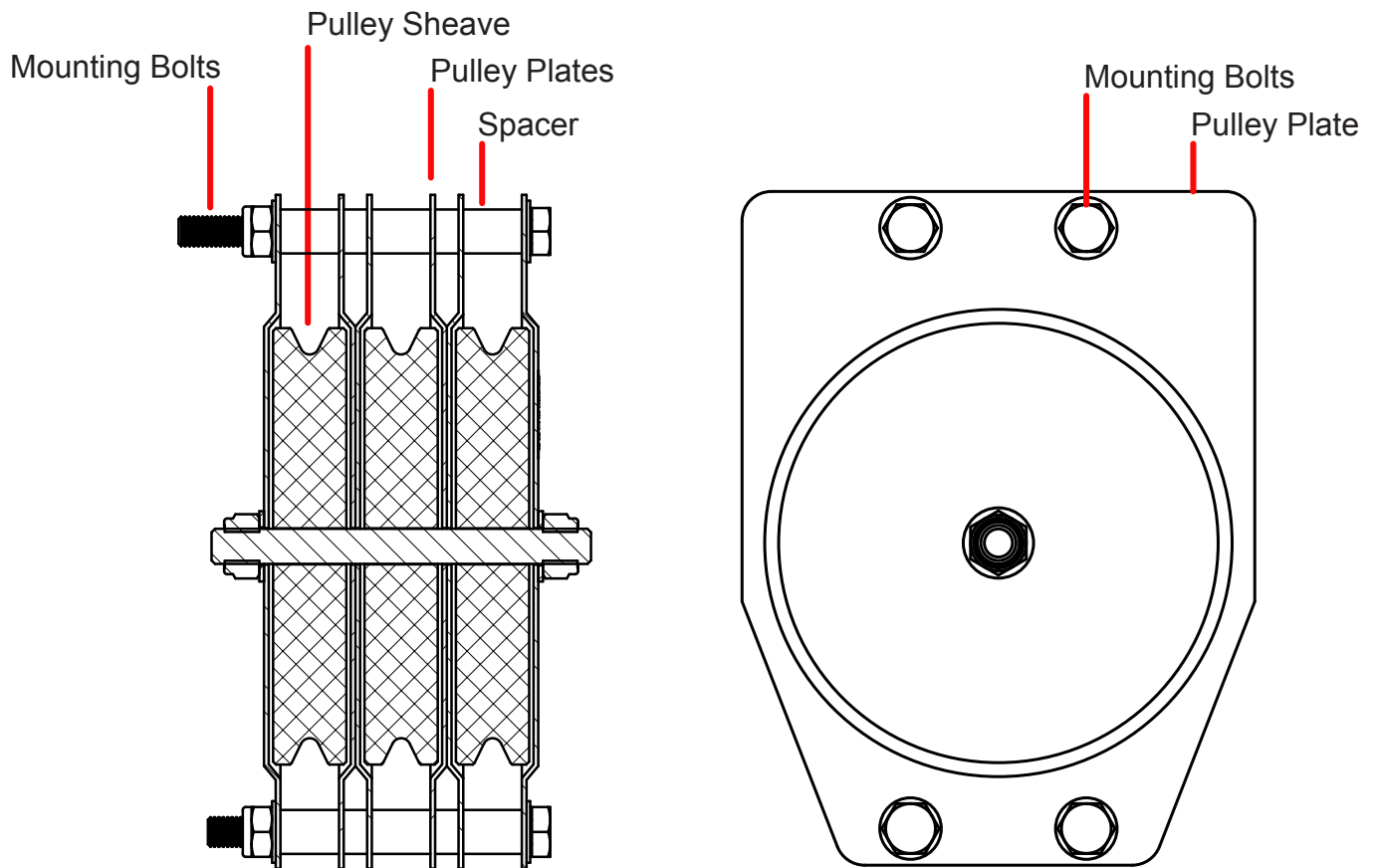
Pt No	Type	Wire/Rope	Max Load	Dia	Width	Weight
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Heavy Duty Pulleys (Wire)

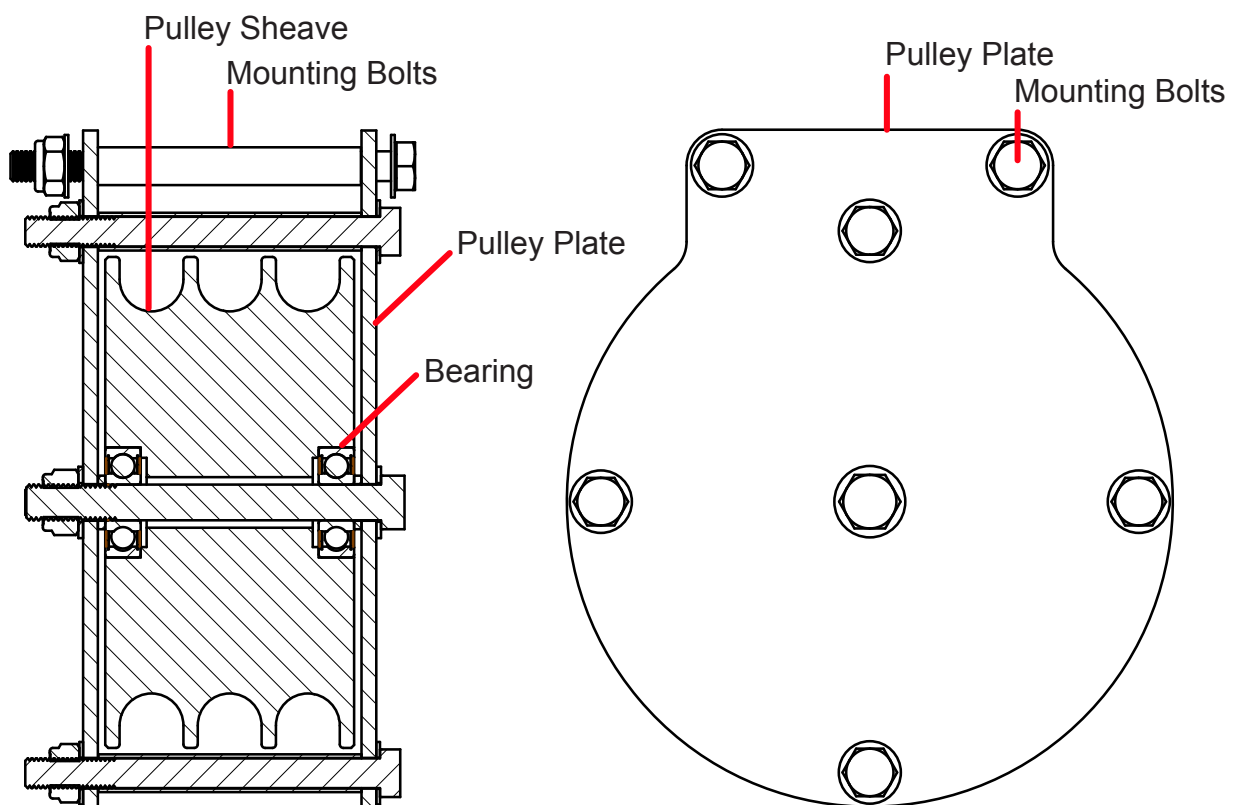
T42901	Single	4mm Wire	200Kg	100mm	50mm	2.80Kg
T42902	Double	4mm Wire	200Kg	100mm	75mm	3.20Kg
T42903	Triple	4mm Wire	200Kg	100mm	100mm	3.60Kg
T42904	Quad	4mm Wire	200Kg	100mm	130mm	4.30Kg
T42905	Quin	4mm Wire	200Kg	100mm	160mm	4.80Kg
T42911	Single	6mm Wire	400Kg	150mm	50mm	4.30Kg
T42912	Double	6mm Wire	400Kg	150mm	75mm	5.00Kg
T42913	Triple	6mm Wire	400Kg	150mm	100mm	5.80Kg
T42914	Quad	6mm Wire	400Kg	150mm	130mm	6.80Kg
T42915	Quin	6mm Wire	400Kg	150mm	160mm	7.60Kg
T42921	Single	6mm Wire	600Kg	200mm	50mm	6.50Kg
T42922	Double	6mm Wire	600Kg	200mm	75mm	7.60Kg
T42923	Triple	6mm Wire	600Kg	200mm	100mm	8.70Kg
T42924	Quad	6mm Wire	600Kg	200mm	130mm	10.40Kg
T42925	Quin	6mm Wire	600Kg	200mm	160mm	11.50Kg

Heavy Duty Pulleys (Rope)

T42931	Single	12mm Rope	200Kg	100mm	50mm	2.70Kg
T42932	Double	12mm Rope	200Kg	100mm	75mm	3.10Kg
T42933	Triple	12mm Rope	200Kg	100mm	100mm	3.50Kg
T42934	Quad	12mm Rope	200Kg	100mm	130mm	4.10Kg
T42935	Quin	12mm Rope	200Kg	100mm	160mm	4.50Kg
T42941	Single	18mm Rope	400Kg	150mm	50mm	4.20Kg
T42942	Double	18mm Rope	400Kg	150mm	75mm	4.80Kg
T42943	Triple	18mm Rope	400Kg	150mm	100mm	5.30Kg
T42944	Quad	18mm Rope	400Kg	150mm	130mm	6.30Kg
T42945	Quin	18mm Rope	400Kg	150mm	160mm	6.90Kg
T42951	Single	18mm Rope	600Kg	200mm	50mm	6.30Kg
T42952	Double	18mm Rope	600Kg	200mm	75mm	7.20Kg
T42953	Triple	18mm Rope	600Kg	200mm	100mm	8.20Kg
T42954	Quad	18mm Rope	600Kg	200mm	130mm	9.70Kg
T42955	Quin	18mm Rope	600Kg	200mm	160mm	10.60Kg



Standard Pulley



Heavy Duty Pulley

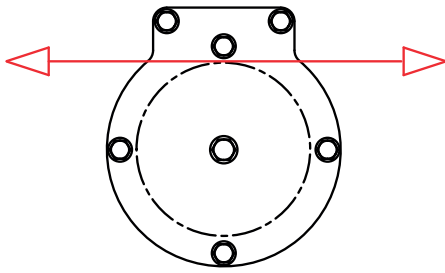
Doughty Pulleys must only be installed/used by qualified/competent personnel.

All Pulleys should be visually inspected before each use. In the event of any damage the pulley should be taken out of service and returned to an 'Approved Doughty Service Agent' for inspection. (See our Dealer Locator on our web site for a comprehensive list of Authorised Service Agents).

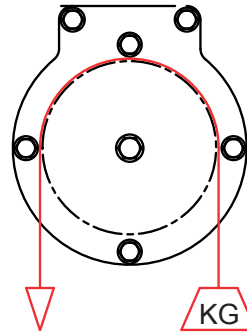
When installing/using Doughty pulleys you must wear appropriate personal protection equipment suitable for the environment you are working in.

Never exceed the Max Load as indicated on the Pulley and in this Manual or Max Load as instructed by your installer, as the installation conditions may alter the Max Load that may applied to the Pulley.

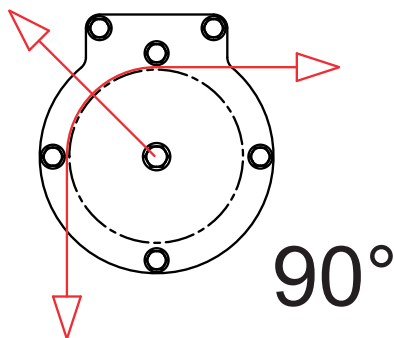
When used in conjunction with any other accessories, care must be taken to ensure the Max Load of any component is not exceeded.



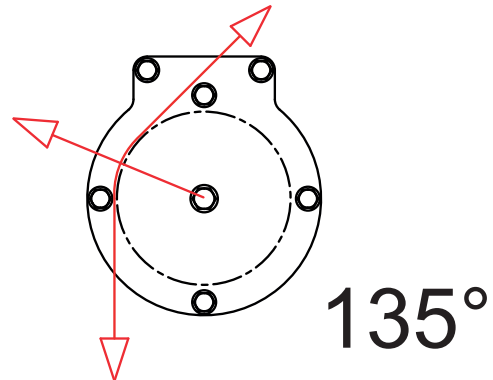
No Load on Pulley



Load on Pulley = $KG \times 2$
ie. Load = 600Kg Actual Load on Pulley = 1200Kg



Load on Pulley = 141% of the Load
ie. Load = 600Kg Actual Load on Pulley = 846Kg



Load on Pulley = 0.76% of the Load
ie. Load = 600Kg Actual Load on Pulley = 456Kg

Care must be taken when installing pulleys to ensure the mounting method can withstand the generated forces

Doughty Engineering stock a range of different mounting methods for these pulleys.

Each mounting supplied by Doughty Engineering carries a SWL to match the pulley to which it will be mounted.

Maintenance

- Maintenance must only be carried out by approved personnel.
- Approved personnel must make regular checks of the product.
- If you see any damage including cracks, deformation or missing parts, the product must be removed from service and returned to an Approved Service Agent for evaluation/repair.
- A full list of Approved Service Agents can be found on our website - www.doughty-engineering.co.uk.
- Approved personnel must carry out a visual and a function test before every usage.
- Approved personnel must carry out more detailed inspection and testing of the product on a regular basis depending on the usage of the product.
- Written records should be kept from the date the product is put into operation.



Warning

- The installer/operator must read and understand the Instruction Manual before using the equipment.
- Operators must be adequately instructed and fully understand the safe use of the equipment before use.
- Failure to carry out maintenance as necessary, including the replacement of parts to the correct standard, could render this equipment unsafe and the manufacturer cannot accept any responsibility in this respect.

Do's and Don'ts

Do ensure the pulley is only used by a competent person

Do ensure the pulley is kept in a clean operational condition

Do check the condition of the pulley regularly

Do use an Authorised Service Agent only for maintenance and repair

Do use a secondary safety

Do Not lift loads which exceed the Max Load of the weakest component in the assembly

Do Not attempt to lift loads which are greater than the Max Load of the pulley

Do Not lift people

Do Not lift loads above people

Do Not lift loads with any personnel in the lifting zone



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Note: Whilst every effort has been made to ensure that the information contained within this manual is correct, Doughty Engineering does not accept any liability for errors or omissions. Specifications and technical data are intended for guidance purposes only and may vary.